

RESEARCH ARTICLE

4EV: Adaptive Video Editing With Spatial Temporal Dynamics and Motion Pathways

NAMRATA PATEL¹, LAKSHMI PRIYA RAMISETTY¹, ADITYA SINGH PARMAR¹,
JIALU LI¹, (Graduate Student Member, IEEE), AND YOUSHAN ZHANG², (Member, IEEE)

¹Artificial Intelligence, Yeshiva University, New York, NY 10033, USA

²School of Artificial Intelligence, Chuzhou University, Chuzhou, Anhui 239000, China

Corresponding author: Youshan Zhang (youshan_zhang@chzu.edu.cn)

ABSTRACT This paper presents 4EV, a novel end-to-end video editing model extending Stable Diffusion and T2I (Text-to-Image) frameworks to generate and edit videos with fine-grained motion dynamics, including object movement, path navigation, background transitions, and zoom effects. 4EV uses spatial-temporal attention for smooth transitions and consistent object appearances, with attention map injection for precise alignment. Contributions include: 1) Motion4EV, a custom dataset of videos and prompts supporting precise motion control, generated through experiments on various motion dynamics to ensure diversity; 2) an enhanced 4EV architecture with attention blocks for improved temporal and spatial refinement; and 3) experimental results demonstrating superior CLIP scores and motion consistency compared to state-of-the-art models. These innovations establish 4EV as a scalable framework for advanced video generation and editing. The source code is available at <https://github.com/npp058/videogen>

INDEX TERMS Video Editing, diffusion models, text-to-video, motion control, one-shot learning, spatial-temporal attention, generative models, zero-shot generation.

I. INTRODUCTION

Video content has become a dominant medium in today's digital era, utilized extensively for storytelling, advertising, and information dissemination. With increasing demands for sophisticated, personalized, and dynamic video experiences, the need for advanced technologies to generate and manipulate videos with high precision has never been more crucial. This paper explores the forefront of video generation technologies, focusing on the intricate customization of motion, text integration, and navigational path dynamics. Leveraging datasets from prior works such as MotionDirector [1], Trailblazer [2], Tune-A-Video [3], and Text2LIVE [4], we aim to develop refined methods that enhance video content's interactivity and engagement by integrating textual prompts and motion dynamics.

Traditional video editing tools are becoming inadequate for modern content creation. For example, Meng and Chang [5] developed a Compressed Video Editing and Parsing System

(CVEPS) with advanced video indexing and manipulation functions. Wang et al. [6] developed an anisotropic kernel mean shift technique to segment the video data into contiguous volumes. Gandhi et al. [7] encoded several common cinematographic practices into a single convex cost function minimization problem. They often lack the flexibility and adaptability necessary for creating real-time, dynamic video content that aligns with evolving user expectations. These tools are limited by their static frameworks, requiring significant manual effort to integrate text, motion, and paths into videos. In contrast, advanced machine learning models offer an innovative approach by enabling adaptive video synthesis that is responsive to user inputs and capable of generating highly customized video experiences.

Our investigation is structured around a series of experiments designed to evaluate and optimize the performance of these state-of-the-art models. We systematically assess how these technologies handle the integration of motion, text, and path customization, identifying key strengths and areas for improvement. We can gain valuable insights into the operational capabilities of each model, laying the groundwork for

The associate editor coordinating the review of this manuscript and approving it for publication was Jolanta Mizera-Pietraszko¹.

transformative enhancements that could revolutionize video generation.

The significance of this study lies in its potential to influence a broad spectrum of industries, from cinematic production and video games to real-time interactive media and virtual reality. Advancing the underlying technology for video customization not only pushes the boundaries of content creation but also supports the development of more engaging, interactive, and tailored video experiences. Each of the models—MotionDirector [1], Trailblazer [2], Tune-A-Video [3], and Text2LIVE [4]—contributes uniquely to the field of video editing and synthesis, offering innovative solutions to the challenges posed by text-guided video manipulation. To summarize, the key contributions of this paper are:

- 1) We propose 4EV, a novel diffusion-based architecture for controllable, high-fidelity text-to-video editing, integrating spatial-temporal attention mechanisms to preserve motion consistency.
- 2) We introduce modular Motion Preservation and Object Injection components with multi-level attention to enable precise semantic editing, trajectory control, and effective text-video alignment.
- 3) We demonstrate that 4EV outperforms existing models by achieving higher CLIP scores for semantic alignment, superior frame consistency for motion smoothness, lower FVD for video quality, and faster generation speed, as shown in our comprehensive quantitative evaluation.

The paper is organized as follows: Section II reviews related work; Section III presents 4EV's architecture; Section IV describes experiments and results; Section V discusses findings; and Section VI concludes the paper.

II. RELATED WORKS

Recent progress in generative modeling has significantly advanced both image and video synthesis from text. Our work builds upon these developments, focusing on spatial-temporal consistency and controllable motion dynamics through attention-based diffusion mechanisms.

A. TEXT-TO-IMAGE (T2I) GENERATION

The evolution of T2I generation has moved from transformer-based models [7], [8] to diffusion models [9], [10], which now dominate the field. Models like GLIDE [11], DALLÉ-2 [12], and Imagen [13] introduced techniques such as classifier-free guidance and CLIP conditioning [14] for improved semantic fidelity. VQdiffusion [15] and Latent Diffusion Models (LDMs) [10] increased computational efficiency by operating in latent spaces. These T2I systems laid the architectural foundation for several text-to-video pipelines.

B. TEXT-TO-VIDEO (T2V) GENERATION

T2V generation extends T2I systems by incorporating temporal coherence between frames. Early works [16], [17] addressed basic motion scenarios, while GODIVA [18], NUWA [19], and MagicVideo [20] added temporal modeling

using sparse attention or multitask training. Imagen Video [13] introduced cascaded video diffusion, and CogVideo [21] enhanced T2I architectures with temporal attention. While powerful in synthesis, these models offer limited interactivity or scene-specific control, and they often require extensive data and computation to generalize well.

C. VIDEO EDITING AND MOTION CONTROL

Text2LIVE [4] enabled texture-level video editing using internal attention maps but struggles with temporal consistency and lacks object-level control. It relies on Layered Neural Atlases [22], which introduces additional computational overhead. Gen-1 [23] and Dreamix [24] introduced more complex motion-aware transformations but rely heavily on large proprietary video datasets and external guidance (e.g., depth, masks). Models such as Control-A-Video [25] and MagicProp [26] introduce structured control through auxiliary signals but require domain-specific tuning.

In contrast, our model 4EV introduces an efficient attention-injection framework that operates in latent space and directly manipulates spatial-temporal representations using cross-attention and temporal attention layers. Unlike Text2LIVE, 4EV preserves motion trajectories and integrates semantic object changes without breaking continuity. Compared to CogVideo, 4EV supports one-shot editing and path-level control with significantly lower training and inference costs, making it more applicable for real-world editing use cases where custom prompts and motion edits are required on minimal data.

D. SINGLE-VIDEO GENERATIVE APPROACHES

Approaches like SinFusion [27] and patch-based Generative Adversarial Networks (GANs) [28] aim to reconstruct new sequences by preserving internal video dynamics but lack the ability to handle semantic changes or text-guided modifications. These methods often require manual configuration and do not scale well to unseen scenarios. In contrast, 4EV maintains semantic alignment and visual coherence while allowing dynamic editing through text prompts, even in zero-shot settings.

4EV therefore provides a unified, low-supervision, and prompt-controllable video editing framework that addresses key limitations in fidelity, interpretability, and usability found in existing methods.

III. METHODS

Our 4EV model leverages DDIM Inversion [29] to encode video into latent space, capturing the essential motion and structure dynamics from the source video. Following this, the model applies motion preservation and object injection techniques, enhanced with multi-level attention blocks (T-Attn, S-Attn) and Feed Forward Network (FFN), to refine both spatial and temporal features.

This architecture enables 4EV to maintain the motion integrity from the original video while adapting specific objects to fit the target prompt. The model incorporates

TABLE 1. Statistics of our collected datasets.

Motion Dynamics	Training (Video + Prompt)	Validation (Prompt)	Testing (Prompt)
Object	42	126	42
Path	16	29	11
Background	12	10	8
Zoom	11	16	8
Total	81	181	69

one-shot video tuning by Wu et al. [3], allowing for high-quality video generation with minimal training data. The denoising step ensures that the output is a polished, high-fidelity video that aligns with the input prompt.

A. DATASET COLLECTION

We curated a dataset comprising 40 distinct videos, selected from a pool of 262 combinations generated using 81 videos and 181 prompts, utilizing MotionDirector [1], Trailblazer [2], Tune-A-Video [3], and Text2LIVE [4] models. The selected videos represent a diverse subset that rigorously evaluates various aspects of video editing capabilities, including motions, colors, trajectories, and programmed zoom actions. This curated subset ensures enhanced diversity, featuring videos with complex motion dynamics, varied backgrounds, and challenging transitions. Furthermore, our evaluation framework emphasizes zero-shot tuning, enabling the model to demonstrate its ability to generalize to unseen scenarios without additional training. By focusing on diversity and zero-shot tuning, this dataset offers a robust benchmark for assessing video synthesis performance.

1) TRAILBLAZER

The Trailblazer [2] model, adapted from the pre-trained ZeroScope T2V model [30], enables precise adjustments to object and character movement paths in videos. Fine-tuned on nearly ten thousand video clips, each consisting of 24 frames, the model allows for precise trajectory modifications. Their model was tested on the Animal Kingdom dataset [31], where the model's ability to follow complex, user-defined routes was evaluated, demonstrating its effectiveness in maintaining narrative continuity seamlessly.

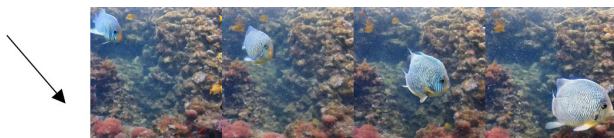


FIGURE 1. In this sequence, a fish's controlled descent from the top left to the bottom right is guided by sequentially placed bounding boxes. This demonstrates precise trajectory manipulation in video synthesis.

2) TUNE-A-VIDEO

The Tune-A-Video model [3] enabled selective transformations and precise modifications in videos, showcasing its potential for dynamic content adaptation. It was trained on

42 videos from the DAVIS dataset, covering a range of categories including animals, vehicles, and humans, allowing the model to effectively learn fine-grained video editing capabilities, as illustrated in Figures 2 and 3.



FIGURE 2. A man is skiing.

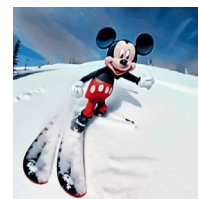


FIGURE 3. A mickey mouse is skiing on the snow.

3) TEXT2LIVE

The Text2LIVE [4] model was utilized to focus on background manipulation across various videos. This model was applied to alter the backgrounds dynamically, integrating diverse scenes to evaluate the model's capability in seamlessly blending new backgrounds while maintaining the integrity of foreground elements. This variety ensures the model's adaptability to changes in scene context without disrupting the primary focus of the video content.



(a) Text prompt: Swarovski blue crystal swan.

FIGURE 4. (a) The retraining efforts have successfully replicated the results outlined in the original study with the same text prompts.

4) MOTIONDIRECTOR

The MotionDirector model [1] was used to manipulate zoom functions in our videos, allowing detailed zoom-ins and broader zoom-outs. It was trained on a dataset of 95 videos featuring 12 human motions, such as playing golf, lifting weights, etc. Each motion had one text prompt, with 72 total prompts used to generate and compare videos.

B. MODEL ARCHITECTURE

In this study, we propose the architecture of our model, 4EV, which bridges the strengths of Text-To-Image (T2I) diffusion



FIGURE 5. Zoom in motion on octopus and dolphin. Row (1) shows the reference video with a zooming motion on a cup. Row (2) displays frames of a dolphin with a zooming motion as well.

models with the temporal dynamics required for text-to-video (T2V) synthesis. Built upon the latent diffusion model (LDM) framework [10], 4EV introduces several architectural enhancements—specifically temporal attention mechanisms, motion preservation, and object injection modules—to support high-quality, coherent, and controllable video generation from textual prompts. The overall pipeline of our approach is illustrated in Figure 6, which shows how a source video is processed, edited, and reconstructed based on textual modifications.

The fundamental challenge in extending T2I models to T2V tasks lies in the temporal dimension that videos introduce. To address this, the models incorporate temporal attention mechanisms, which are crucial for maintaining continuity across frames, allowing the model to generate coherent sequences over time. The models also incorporate Frame Prediction Modules integrated within the architecture; these modules predict subsequent frames based on both the current frame and the textual description, ensuring dynamic consistency throughout the video.

The T2I diffusion model, for instance, LDM [10], primarily utilizes a U-Net structure [32]. This structure involves an initial phase of spatial downsampling, followed by a phase of upsampling, linked through skip connections. It is built with layers consisting of 2D convolutional residual blocks and transformer blocks. Each transformer block contains a spatial self-attention layer that detects correlations by examining pixel locations within feature maps, a cross-attention layer that assesses the relationship between pixels and contextual inputs like text, and a Feed Forward Network (FFN).

DDIM (Denoising Diffusion Implicit Models) Inversion [29]: The transformation initiates with the DDIM inversion process, where the input video V is encoded into a latent space representation Z_T . The text-to-video diffusion model is trained to generate video frames that align with a given textual description. The training process utilizes a pair of inputs: a video V with frames $\{v_1, v_2, \dots, v_n\}$ and a text description T . The model is trained to minimize the difference between the generated frames and the original video frames.

1) TRANSFORMATION MODULES

Z_T represents the latent representation of the image at a particular timestep. In diffusion models, images are progressively denoised from a noisy state to a clean state, and Z_T symbolizes the state of the image at time t . Z_0 represents the latent state of the image with the lowest level of noise, closest to the final clean image.

a: MOTION PRESERVATION MODULE

This module is designed to maintain the intrinsic motion captured in the latent space Z_T . Using a combination of Spatial-Temporal Attention (ST-Attn) and Temporal Attention (T-Attn), the motion dynamics are preserved. For example, the motion preservation ensures the car keeps moving in the same direction and manner.

b: OBJECT INJECTION MODULE

The object injection modifies the object, like changing a rabbit to a dog without affecting its trajectory or speed. Z_T is processed to alter or introduce new elements per the target prompt.

c: STRUCTURE CONTROL

This step refines the changes made during the object injection phase, ensuring the structural integrity of the video is maintained (e.g., the background still looks the same). Both Z_0^r and Z_0^m are integrated using a structure control function that optimizes coherence and alignment with the target prompt.

2) MODEL COMPONENTS

For each block of the denoising UNet, there are four basic modules: a convolutional module, a spatial-temporal self-attention module (ST-Attn), a spatial cross-attention module (CA-S), a Feed Forward Network (FFN), and a temporal self-attention module (T-Attn).

a: ATTENTION BLOCKS

- **Spatial-Temporal Attention (ST-Attn):** This mechanism focuses on capturing dependencies across different frames and within frames over time, enhancing the model's ability to understand motion and changes in the scene.
- **Cross-Attention (CA-S):** It allows the model to focus on specific parts of the text and video frames that are relevant to each other, facilitating a better alignment of the video content with the text description.
- **Temporal Attention (T-Attn):** This mechanism focuses on the temporal relationships between video frames, ensuring that transformations maintain natural continuity and timing. T-Attn preserves the narrative flow and prevents abrupt or unrealistic transitions. This helps ensure that the video remains coherent and immersive after transformation.

To enable fine-grained modeling, we adapt the spatial self-attention mechanism by incorporating spatial-temporal

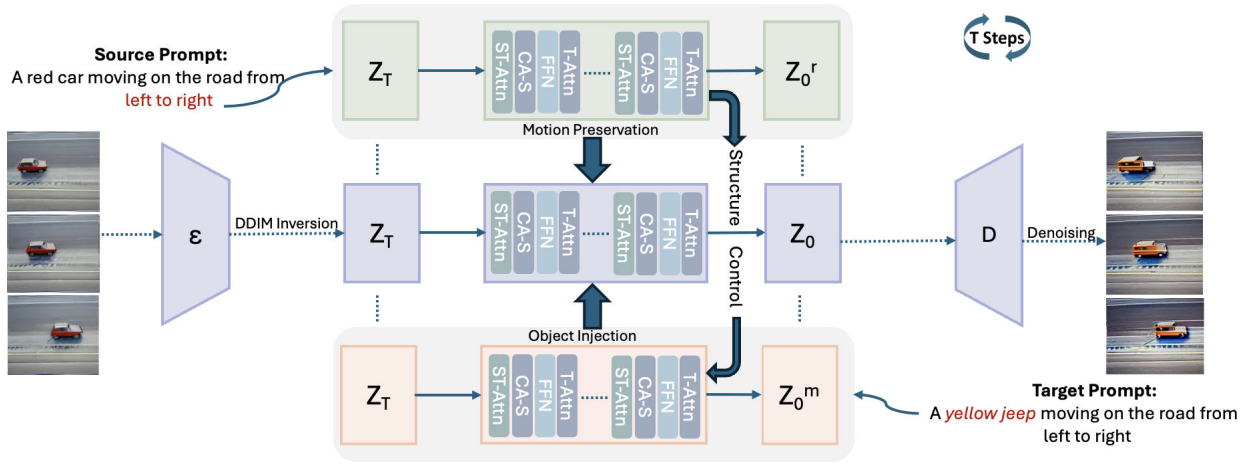


FIGURE 6. 4EV Pipeline starting with a source prompt (“A red car moving on the road from left to right”) and corresponding video, the video is first encoded into a latent representation through DDIM Inversion. It then undergoes motion preservation and object injection using attention mechanisms and Feed Forward Network (FFN) to adjust according to the target prompt (“A yellow jeep moving on the road from left to right”). The combined latent representation is refined through a denoising process to produce the final high-quality video output.

self-attention across frames, which can be formulated as:

$$\text{SelfAttn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (1)$$

where

$$\begin{aligned} Q &= W^Q \bar{v}_i, \\ K &= W^K [\bar{v}_0, \dots, \bar{v}_{N-1}], \\ V &= W^V [\bar{v}_0, \dots, \bar{v}_{N-1}]. \end{aligned}$$

where $\bar{v}_i$ denotes the token sequence of frame i , and $[\bar{v}_0, \dots, \bar{v}_{N-1}]$ denotes the concatenation of the N frames. We concatenate features K, V of N frames so that each position has a global perception of all video frames and tends to generate more consistent results.

b: FEED FORWARD NETWORK (FFN)

The FFN acts as the final processing stage in our model, where it refines the outputs from the attention mechanisms to enhance frame reconstruction. It adjusts critical features such as detail and color accuracy, ensuring the produced frames are realistically portrayed.

In video editing, Feed Forward Networks (FFNs) are useful for tasks like scene classification, object recognition, and video generation. They model relationships within and between frames by approximating complex functions. Learning involves adjusting network parameters to minimize error, typically using backpropagation and optimization methods like Stochastic Gradient Descent (SGD) for efficient learning on large video datasets.

c: RECONSTRUCTION LOSS

We evaluate the model’s performance using the reconstruction loss, which measures the discrepancy between the original frames and those predicted by the model. This loss is primarily calculated using Mean Squared Error (MSE),

which quantifies the average squared difference between the original and predicted pixel values. A lower MSE indicates a higher fidelity in video reconstruction, which is critical for the realism and utility of the transformed video content.

$$\text{Loss} = \frac{1}{n} \sum_{i=1}^n \|\hat{v}_i - v_i\|^2 \quad (2)$$

The training involves iteratively updating the weights of the non-frozen parts of the model (mainly the attention blocks and Feed Forward Network (FFN)) to minimize the reconstruction loss. This iterative process gradually improves the model’s ability to generate accurate and coherent video frames from textual descriptions.

d: DENOISING AND ZERO-SHOT TUNING

In our framework, denoising is performed via DDIM (Denoising Diffusion Implicit Models) inversion, enabling fast and deterministic reconstruction of clean frames from noisy latent representations for controllable video editing. Operating in a zero-shot setting with a single short video (24–48 frames) as input, the base diffusion model remains frozen while only selected attention modules (ST-Attn, T-Attn, and CA-S) are updated, allowing the system to preserve motion dynamics and generalize to new prompts without extensive retraining or large datasets.

C. ENHANCED END-TO-END VIDEO EDITING

To address the complexity of text-guided video editing, our model integrates advanced attention modules designed to refine both spatial and temporal features. The Motion Preservation Module utilizes Temporal Attention (T-Attn) and Spatial-Temporal Self-Attention (ST-Attn) to preserve object motion dynamics, including speed, direction, and smoothness, thereby ensuring temporal coherence across frames. The Object Injection Module modifies objects in the latent

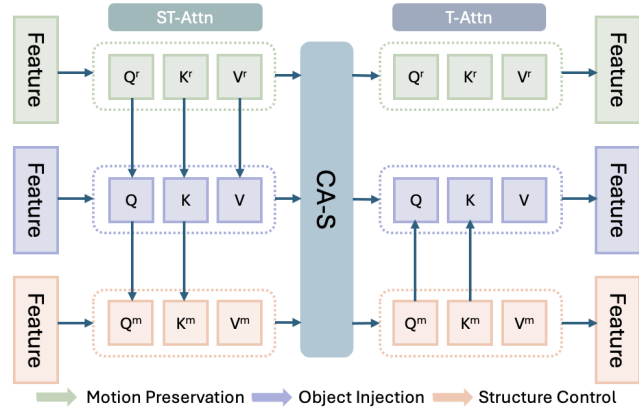


FIGURE 7. Integration of spatial-temporal attention (ST-attn) and temporal attention (T-attn) within the cross attention Stage (CAS) framework. This enables dynamic video editing by preserving motion trajectories, enabling object transformations, and maintaining spatial structure through the coordinated use of query (Q), key (K), and value (V) features across time and space.

space using Cross-Attention (CA-S) to align textual prompts with latent tokens, enabling object replacement without altering motion trajectories. Finally, Multi-Level Attention Blocks combine ST-Attn, T-Attn, and CA-S within each U-Net block, capturing both intra- and inter-frame dependencies for coherent text-guided synthesis.

By recursively refining latent representations through these mechanisms, the model produces temporally stable and semantically accurate video edits. This design is particularly effective in real-time and controllable video editing tasks, where dynamic elements must remain coherent throughout the sequence.

The **Spatial-Temporal Attention** mechanism jointly processes spatial and temporal information, ensuring seamless transitions and consistent object appearances across frames. In contrast, **Temporal Attention** focuses primarily on frame-to-frame continuity, preserving motion integrity in dynamic scenes with continuous object movement.

We define the attention map injection on temporal self-attention layers as follows:

$$T\text{-Attn}^l := \begin{cases} \text{SelfAttn}(Q^m, K^m, V), & \text{if } t > t_1 \text{ and } l > l_1 \\ \text{SelfAttn}(Q, K, V), & \text{otherwise} \end{cases} \quad (3)$$

where Q^m and K^m refer to the query and key of the object injection branch. In practice, we set $t_1 = 0$ and $l_1 = 0$. The injection of temporal attention maps has been empirically shown to improve alignment of generated objects with the target prompt, enabling more controlled and consistent editing outcomes.

D. EVALUATION

To evaluate the quality and relevance of generated videos, we assess four key attributes: semantic alignment, temporal coherence, overall video quality, and generation efficiency.

Semantic alignment is measured using CLIP Score, computing the cosine similarity between generated frames and input prompts via CLIP embeddings [14], where higher scores indicate stronger text-video alignment. Temporal coherence is assessed via Frame Consistency, capturing how smoothly objects move across frames. Overall video quality is evaluated using FVD (Fr chet Video Distance), with lower scores indicating higher realism and temporal consistency. Latency (Runtime) is reported as average generation time per frame, serving as a proxy for real-time feasibility.

Our evaluation uses a curated benchmark of 181 text prompts paired with 81 source videos, categorized into object, path, zoom, and background edits. Each model is evaluated on the same subset to ensure consistent comparison. However, it is important to note that while baseline models support individual edits (e.g., object injection in Text2LIVE or zoom edits in MagicVideo), they do not support simultaneous multi-attribute edits within a single framework. In contrast, 4EV enables combined edits such as object replacement with motion trajectory modification and zoom transformations using a single unified prompt.

TABLE 2. Quantitative comparison with state-of-the-art models.

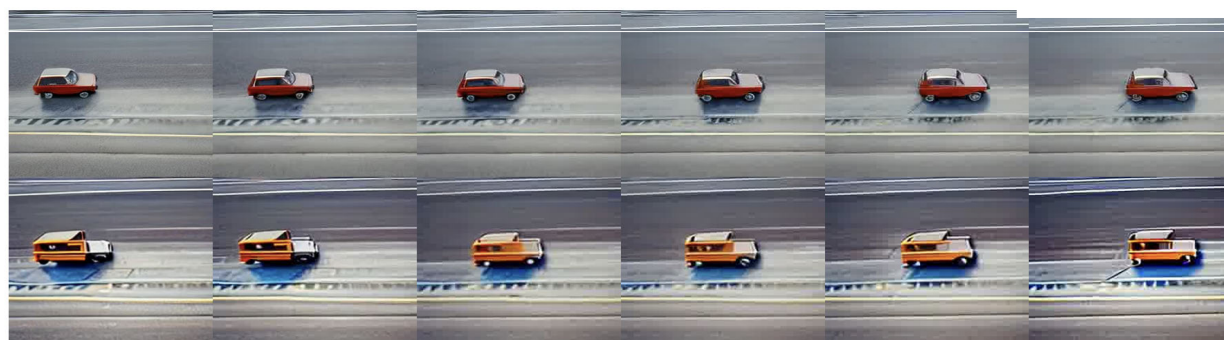
Method	CLIP	Consistency	FVD	Latency (s/f)
MagicVideo [5]	0.87	0.68	250	2.4
Plug-And-Play [15]	0.88	0.69	240	2.6
CogVideo [21]	0.90	0.71	220	2.8
Text2LIVE [4]	0.88	0.67	230	2.3
Dreamix [24]	0.91	0.72	210	2.1
4EV (Ours)	0.92	0.79	180	1.8

As shown in Table 2, 4EV achieves the highest CLIP Score, indicating stronger alignment between visual content and textual prompts, and the lowest FVD score (180), suggesting superior overall video quality and temporal consistency. These results confirm the model’s ability to generate coherent, high-fidelity video edits with minimal delay.

TABLE 3. Evaluation results for some of our outputs.

Video	CLIP	Consist.	Edit Type(s)	Success (%)
Yellow jeep moving left to right	0.93	0.98	Object, Motion	95%
Dog walking top left to bottom right	0.92	0.98	Object, Motion	90%
Tortoise swimming right to left	0.91	0.91	Object, Motion	94%
Tiger galloping right to left	0.90	0.93	Object, Motion	88%
Fish diving underwater right to left (zoom in)	0.88	0.92	Object, Motion, Zoom	80%
Average	0.91	0.94	–	89%

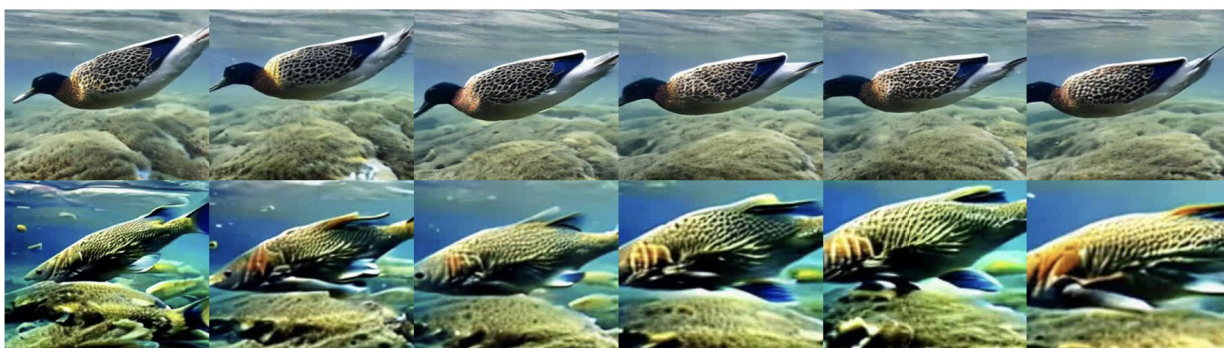
Table 3 reports CLIP Scores (semantic similarity), Frame Consistency (temporal coherence), Edit Types, and Success



A *red car* moving on the road from *left to right*



A *yellow jeep* moving on the road from *left to right*



A *duck* diving underwater from *right to left*



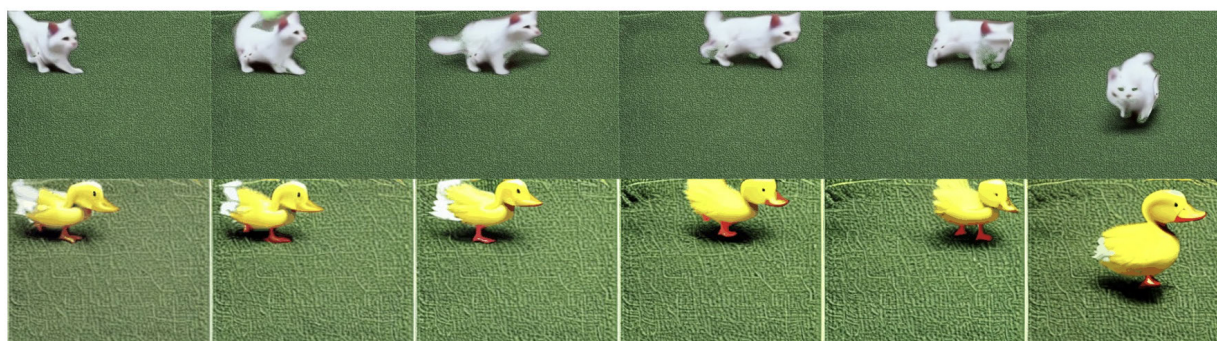
A *fish* diving underwater from *right to left* with a *zoom in*



A *rabbit* walking on ground from *top left to bottom right*



A *dog* walking on ground from *top left to bottom right*



A *white cat* walking on grass from *top left to top right then down*



A *yellow duck* walking on grass from *top left to top right then down*

FIGURE 8. Generated video of car to jeep, duck to fish, rabbit to dog and cat to duck.

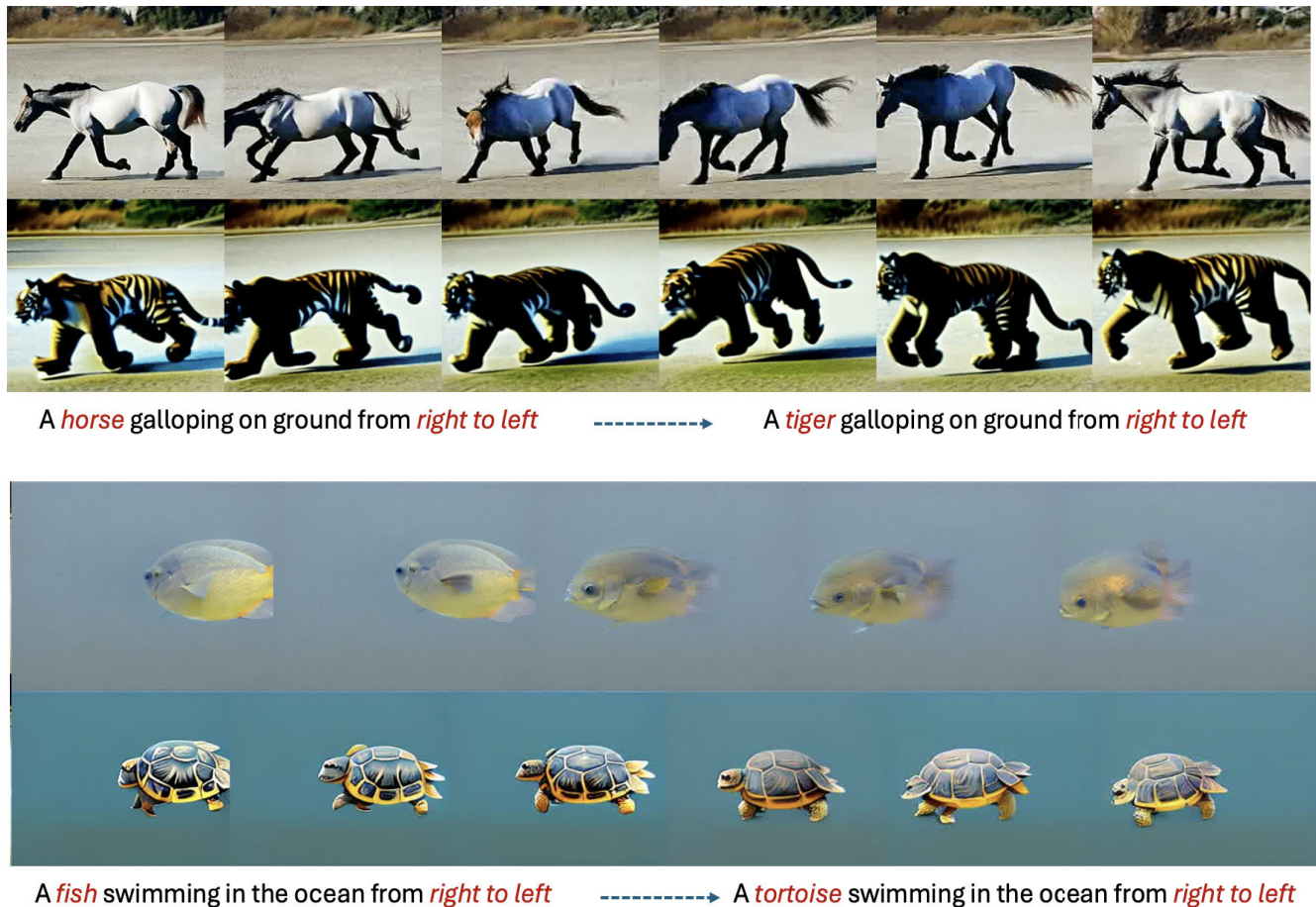


FIGURE 9. More generated video of horse to tiger and fish to tortoise.

Percentages (subjective effectiveness ratings) for selected outputs. For example, the video of a yellow jeep achieved a CLIP Score of 0.93, frame consistency of 0.98, and a 95% success rating for object and motion edits, indicating strong text alignment, smooth motion, and effective editing. The tortoise example achieved a CLIP Score of 0.91 and a 94% success rating, reflecting high semantic fidelity with effective motion editing. Overall, the average success rate of 89% across these diverse edit types demonstrates that the model produces high-quality, semantically aligned, and visually coherent videos suitable for practical applications.

IV. RESULTS

Table 1 presents the dataset statistics employed in our evaluation, outlining the diversity and scope of the video samples. Table 2 compares the performance of 4EV with baseline methods across standard evaluation metrics, while Table 3 provides a detailed breakdown of results across different types of video edits.

The qualitative results illustrated in Figures 8 and 9 highlight the model's effectiveness in capturing realistic motion dynamics, trajectory alignment, and spatial zoom effects. For example, vehicles such as cars and jeeps follow user-defined paths with high precision, while aquatic creatures like ducks

and fish demonstrate smooth, naturalistic swimming behavior enhanced by dynamic perspective zooms. Terrestrial animals, including rabbits and dogs, exhibit multi-directional locomotion, and domestic subjects like a white cat or a yellow duck perform complex, prompt-guided transitions with fine-grained motion coherence. These experiments collectively demonstrate the model's capability to generalize across diverse environments while preserving semantic and temporal consistency.

V. DISCUSSION

This study presents a novel video synthesis framework, 4EV, that dynamically edits motion path, object, background, and zoom effects in response to textual prompts. By combining spatial-temporal attention, motion preservation, and object injection, 4EV provides fine-grained control over video generation, enabling semantically aligned and temporally consistent outputs with minimal supervision.

A. ADVANCEMENTS AND APPLICATIONS

The ability to manipulate multiple video attributes simultaneously introduces new potential in automated content creation. In virtual reality, 4EV could support the dynamic generation of immersive environments responsive to real-time input.

In education, it could create tailored visualizations to match specific curricula. In media and game production, 4EV can accelerate previsualization and post-production by enabling prompt-driven, localized edits without retraining.

B. PRACTICAL FEASIBILITY AND DEPLOYMENT

We evaluated 4EV on a single NVIDIA A100 GPU. The model generates 512×512 frames in 1.5–2.2 seconds each, depending on scene complexity and motion prompts. While not fully real-time, this latency is suitable for near-real-time applications such as rapid content iteration in creative workflows or interactive design tools. Inference requires 18–22 GB of VRAM, making it compatible with high-end GPUs such as the RTX 3090. Future work will focus on reducing inference costs through model distillation and memory-efficient optimization strategies.

Figures 8 and 9 qualitatively illustrate 4EV's ability to preserve motion and semantic alignment across diverse scenarios, including vehicle paths, aquatic motion, and animal locomotion. The system currently assumes clean source videos and performs best on short sequences with well-defined motion. Challenges remain in scaling to longer videos, handling occlusions, and performing compound edits involving multiple moving elements.

C. COMPARATIVE ANALYSIS WITH PRIOR MODELS

Compared to prior works like Text2LIVE and CogVideo, 4EV demonstrates clear advantages in editability, motion control, and semantic fidelity. Text2LIVE relies on layered neural atlases and is limited to texture-level edits with reduced motion consistency. CogVideo enhances a T2I model with temporal attention but lacks localized object control and precise motion alignment. In contrast, 4EV integrates attention map injection and multi-stage spatial-temporal attention, enabling path control, object substitution, and zoom-based transformations from a single prompt.

As shown in Table 2, 4EV achieves the highest CLIP score (0.92) and improved temporal consistency over all compared models, while maintaining efficient inference performance. These attributes make it a flexible and competitive solution for prompt-driven video editing across domains.

D. CHALLENGES AND LIMITATIONS

Despite its strengths, 4EV faces ongoing challenges. Capturing a wide range of real-world motion dynamics in a benchmark dataset remains difficult. Ensuring robust generation in videos with blur, occlusion, or erratic movement also remains an open problem. Technically, aligning object injection with fine-grained attention while maintaining motion integrity was a significant challenge during development and will continue to be an area of optimization.

Overall, 4EV takes an important step toward generalizable, high-quality, and controllable video editing while maintaining competitive performance across semantic, temporal, and computational dimensions.

VI. CONCLUSION

The research introduces a transformative approach to video generation that integrates text-to-video synthesis capabilities and the flexibility to modify key visual components in real time. The study shows the effectiveness of the 4EV model in changing paths, objects, backgrounds, and motion within videos based on textual descriptions. This capability marks a substantial improvement in generative video methods, offering more nuanced and controlled video editing that aligns closely with human input and creative direction. Future research will focus on optimizing the model's computational efficiency and improving its ability to handle longer sequences with more complex scenarios.

REFERENCES

- [1] R. Zhao, Y. Gu, J. Z. Wu, D. J. Zhang, J. Liu, W. Wu, J. Keppo, and M. Z. Shou, "MotionDirector: Motion customization of text-to-video diffusion models," 2023, *arXiv:2310.08465*.
- [2] W.-D. K. Ma, J. P. Lewis, and W. B. Kleijn, "TrailBlazer: Trajectory control for diffusion-based video generation," 2023, *arXiv:2401.00896*.
- [3] J. Z. Wu, Y. Ge, X. Wang, W. Lei, Y. Gu, Y. Shi, W. Hsu, Y. Shan, X. Qie, and M. Z. Shou, "Tune-A-Video: One-shot tuning of image diffusion models for text-to-video generation," 2022, *arXiv:2212.11565*.
- [4] O. Bar-Tal, D. Ofri-Amar, R. Fridman, Y. Kasten, and T. Dekel, "Text2LIVE: Text-driven layered image and video editing," 2022, *arXiv:2204.02491*.
- [5] J. Meng and S.-F. Chang, "CVEPS—A compressed video editing and parsing system," in *Proc. 4th ACM Int. Conf. Multimedia*, Seattle, WA, USA, 1997, pp. 43–53.
- [6] Z. Wang, A. Li, L. Zhu, Y. Guo, Q. Dou, and Z. Li, "CustomVideo: Customizing text-to-video generation with multiple subjects," 2024, *arXiv:2401.09962*.
- [7] V. Gandhi, R. Ronfard, and M. Gleicher, "Multi-clip video editing from a single viewpoint," in *Proc. 11th Eur. Conf. Vis. Media Prod.*, Nov. 2014, pp. 1–10.
- [8] J. Yu, X. Li, J. Y. Koh, H. Zhang, R. Pang, J. Qin, A. Ku, Y. Xu, J. Baldridge, and Y. Wu, "Vector-quantized image modeling with improved VQGAN," 2021, *arXiv:2110.04627*.
- [9] J. Sohl-Dickstein, E. A. Weiss, N. Maheswaranathan, and S. Ganguli, "Deep unsupervised learning using nonequilibrium thermodynamics," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 2256–2265.
- [10] L. Weng. (2021). *What Are Diffusion Models?*. Accessed: Jun. 25, 2025. [Online]. Available: <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>
- [11] J. Wang, Y. Xu, H.-Y. Shum, and M. F. Cohen, "Video tooning," in *Proc. ACM SIGGRAPH Papers*, Aug. 2004, pp. 574–583.
- [12] J. Yu, Y. Xu, J. Yu Koh, T. Luong, G. Baid, Z. Wang, V. Vasudevan, A. Ku, Y. Yang, B. K. Ayan, B. Hutchinson, W. Han, Z. Parekh, X. Li, H. Zhang, J. Baldridge, and Y. Wu, "Scaling autoregressive models for content-rich text-to-image generation," 2022, *arXiv:2206.10789*.
- [13] J. Ho, W. Chan, C. Saharia, J. Whang, R. Gao, A. Gritsenko, D. P. Kingma, B. Poole, M. Norouzi, D. J. Fleet, and T. Salimans, "Imagen video: High definition video generation with diffusion models," 2022, *arXiv:2210.02303*.
- [14] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," 2021, *arXiv:2103.00020*.
- [15] M. Ku, C. Wei, W. Ren, H. Yang, and W. Chen, "AnyV2V: A tuning-free framework for any video-to-video editing tasks," 2024, *arXiv:2403.14468*.
- [16] N. Haim, B. Feinstein, N. Granot, A. Shocher, S. Bagon, T. Dekel, and M. Irani, "Diverse generation from a single video made possible," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 491–509.
- [17] C. Meng, Y. He, Y. Song, J. Song, J. Wu, J.-Y. Zhu, and S. Ermon, "SDEdit: Guided image synthesis and editing with stochastic differential equations," in *Proc. Int. Conf. Learn. Represent.*, 2021.
- [18] C. Zhang, C. Zhang, M. Zhang, I. S. Kweon, and J. Kim, "Text-to-image diffusion models in generative AI: A survey," 2023, *arXiv:2303.07909*.

- [19] C. Wu, J. Liang, L. Ji, F. Yang, Y. Fang, D. Jiang, and N. Duan, "Nuwa: Visual synthesis pre-training for neural visual world creation," in *Proc. ECCV*, 2022, pp. 720–736.
- [20] O. Gafni, A. Polyak, O. Ashual, S. Sheynin, D. Parikh, and Y. Taigman, "Make-A-scene: Scene-based text-to-image generation with human priors," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*. Tel Aviv, Israel: Springer, 2022, pp. 89–106.
- [21] W. Hong, M. Ding, W. Zheng, X. Liu, and J. Tang, "CogVideo: Large-scale pretraining for text-to-video generation via transformers," 2022, *arXiv:2205.15868*.
- [22] Y. Kasten, D. Ofri, O. Wang, and T. Dekel, "Layered neural atlases for consistent video editing," *ACM Trans. Graph.*, vol. 40, no. 6, pp. 1–12, Dec. 2021.
- [23] P. Esser, J. Chiu, P. Atighehchian, J. Granskog, and A. Germanidis, "Structure and content-guided video synthesis with diffusion models," 2023, *arXiv:2302.03011*.
- [24] E. Molad, E. Horwitz, D. Valevski, A. R. Acha, Y. Matias, Y. Pritch, Y. Leviathan, and Y. Hoshen, "Dreamix: Video diffusion models are general video editors," 2023, *arXiv:2302.01329*.
- [25] W. Chen, Y. Ji, J. Wu, H. Wu, P. Xie, J. Li, X. Xia, X. Xiao, and L. Lin, "Control-A-video: Controllable text-to-video diffusion models with motion prior and reward feedback learning," 2023, *arXiv:2305.13840*.
- [26] H. Yan, J. H. Liew, L. Mai, S. Lin, and J. Feng, "MagicProp: Diffusion-based video editing via motion-aware appearance propagation," 2023, *arXiv:2309.00908*.
- [27] Y. Nikankin, N. Haim, and M. Irani, "SinFusion: Training diffusion models on a single image or video," 2022, *arXiv:2211.11743*.
- [28] S. Gur, S. Benaim, and L. Wolf, "Hierarchical patch VAE-GAN: Generating diverse videos from a single sample," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 16761–16772.
- [29] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," 2020, *arXiv:2010.02502*.
- [30] C. Qi, X. Cun, Y. Zhang, C. Lei, X. Wang, Y. Shan, and Q. Chen, "FateZero: Fusing attentions for zero-shot text-based video editing," 2023, *arXiv:2303.09535*.
- [31] X. L. Ng, K. E. Ong, Q. Zheng, Y. Ni, S. Y. Yeo, and J. Liu, "Animal kingdom: A large and diverse dataset for animal behavior understanding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 19001–19012.
- [32] X. Zhang and M. Agrawala, "Controlnet: Conditional control of diffusion models," 2023, *arXiv:2302.05543*.



NAMRATA PATEL received the M.S. degree in artificial intelligence and extensive experience in software and full-stack engineering. She is currently pursuing the Ph.D. degree with Yeshiva University. She is currently a Graduate Research Assistant with Yeshiva University, specializing in artificial intelligence research and development. She focuses on bridging AI research with production-grade systems. Her work spans generative AI, large language models (LLMs), computer vision, deep learning, and retrieval-augmented generation (RAG), emphasizing the transition from theoretical concepts to deployable AI solutions.



LAKSHMI PRIYA RAMISETTY is a Researcher and Engineer specializing in machine learning, large language models (LLMs), and scalable AI-driven systems. Her work spans generative AI, information retrieval, and multimodal learning, with expertise in fine-tuning language models, optimizing neural architectures, and developing large-scale data pipelines. She has built real-time fraud detection systems, conversational AI applications, and deep learning-based image processing frameworks. With hands-on experience in cloud-based ML solutions, including GCP and Vertex AI, she focuses on enhancing model efficiency and scalability in practical AI applications.



ADITYA SINGH PARMAR received the bachelor's degree from L.D. College of Engineering (GTU), Gujarat, India, and the M.S. degree in artificial intelligence from Yeshiva University. He is currently an AI Researcher and Engineer with industry experience in machine learning, computer vision, and edge AI. His experience and interests include developing innovative solutions using large language models (LLMs), deep learning, reinforcement learning, Edge AI, video finetuning, and the IoT.



JIALU LI (Graduate Student Member, IEEE) is currently pursuing the master's degree in artificial intelligence with Yeshiva University. She is a Student Research Assistant with the Department of Graduate Computer Science and Engineering. She is enthusiastic about interdisciplinary research and devoted to the practical application of generative AI and other state-of-the-art technologies. Her research interests include deep learning, computer vision, and multimodal learning.



YOUSHAN ZHANG (Member, IEEE) received the Ph.D. degree in computer science from Lehigh University. He was an Assistant Professor with the Department of Graduate Computer Science and Engineering, Yeshiva University. He was a Postdoctoral Researcher with Cornell University. He is currently a Professor with the School of Artificial Intelligence, Chuzhou University. His research involves artificial intelligence, machine learning, computer vision, transfer learning, manifold learning, and shape analysis. Specifically, he focuses on image, audio, and video classification, segmentation, regression, generation, as well as visual language models (VLMs) development.

...